

Autonomous Contextual Authentication for Web Applications Combining an AI Ensemble and Permissioned Blockchain Auditing

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Abstract—Traditional authentication mechanisms based on static passwords or fixed multi-factor authentication (MFA) remain widely deployed, yet they ignore the behavioral and environmental context of each access attempt and are increasingly exploited by credential stuffing, phishing, and account-takeover attacks. This paper presents an autonomous contextual authentication system for web applications that combines a machine-learning ensemble with permissioned-blockchain auditing. Each login is described by fourteen contextual and behavioral features, which are scored by an ensemble of Isolation Forest, Random Forest, and a multilayer neural network; the resulting risk score drives a three-band adaptive decision (grant, step-up to MFA, or block). Every decision is recorded immutably on a Hyperledger Fabric ledger, yielding a verifiable, non-repudiable audit trail whose necessity over signed logs and append-only databases is explicitly justified. In a controlled evaluation on a synthetic dataset of about 50,000 login records (95% legitimate, 5% suspicious), the ensemble reached 98.1% accuracy, 80.2% F1-score, and a 0.99 ROC-AUC on a held-out test set, with a 0.8% false-positive rate and an inference latency of roughly 33 ms per request. These results are preliminary and rely on synthetic data; they characterize the technical feasibility of the approach rather than a field study, and are to be confirmed with real data.

Index Terms—contextual authentication, risk-based authentication, machine learning, ensemble, anomaly detection, blockchain, audit trail, information security.

I. INTRODUCTION

Authentication is the first line of defense of web applications. Historically it has relied on a knowledge factor—the password—whose weaknesses are well documented: passwords are reused across services, leaked at scale, and exploited through brute force, credential stuffing, and phishing [1], [2]. Multi-factor authentication (MFA) mitigates part of this exposure by combining knowledge, possession, and inherence factors, but static MFA still falls to social engineering, SIM swapping, and real-time code interception, while adding friction to the user experience [1].

The scale of the problem justifies moving beyond the credential alone. Credential stuffing, phishing, and the use of stolen credentials remain among the leading vectors in breach reports, and account-takeover fraud imposes direct financial and reputational costs on organizations and end users [2]. Password reuse turns any single leak into a cross-service liability, while commoditized attack tooling makes large-scale login abuse cheap and highly automated. Crucially, static defenses—password-strength policies and one-time codes—do not observe *how*, *when*, or *from where* a login occurs, and therefore cannot distinguish a legitimate user from an adversary who already holds valid credentials. Regulatory frameworks such as the Brazilian LGPD [21] add a further requirement: automated access decisions over personal data must be demonstrably accountable. Together, these pressures motivate authentication decisions that are conditioned on the full context of each attempt—and that leave a verifiable trail of *why* each decision was taken.

To address these limitations, the NIST digital identity guidelines and the Zero Trust model recommend *adaptive* mechanisms, in which the level of verification required is proportional to the estimated risk of each attempt [1], [3]. This shifts authentication from a binary, static decision toward a continuous, context-sensitive assessment—an approach known in the literature as risk-based authentication (RBA). Empirical studies of production services show, however, that deployed RBA is dominated by a single signal (typically the IP address) and by hand-tuned heuristics [4], [5]. In parallel, when access decisions are delegated to autonomous models, they must be *auditable* by third parties (e.g., for regulatory compliance), a requirement that decision logs held by a single operator do not fully satisfy.

This paper proposes an autonomous contextual authentication system that unifies, in a single flow, an adaptive machine-

learning (ML) decision and its immutable audit. The main contributions are:

- an end-to-end architecture that fuses fourteen contextual and behavioral features into a single risk score computed by an ensemble of Isolation Forest, Random Forest, and a neural network, driving a three-band adaptive decision;
- a permissioned-blockchain audit layer coupled to the decision, with an explicit justification of *why* a blockchain is needed over signed logs or append-only databases; and
- a reproducible evaluation methodology, with baselines and a preliminary feasibility study on synthetic data.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the proposed architecture, Section IV the AI decision engine, and Section V the blockchain audit layer. Section VI presents the evaluation methodology and Section VII the results. Section VIII concludes.

II. RELATED WORK

Behavioral biometrics. Continuous authentication from interaction patterns is well established. TypeNet [11] and TypeFormer [12] identify users from keystroke dynamics with low error rates, and Touchalytics [13] shows that touchscreen patterns discriminate users on mobile devices. These works confirm the discriminative power of user behavior but focus on a single biometric modality and do not address immutable auditing of the decisions. Contemporary surveys of sensor-based continuous authentication [9] corroborate that behavioral signals generalize across many modalities, while highlighting open challenges in robustness, template ageing, and real-world deployment—challenges that argue for fusing behavior with other contextual signals rather than relying on a single modality.

Contextual and device signals. Beyond behavior, the device and network context of a login is itself discriminative. Alaca and van Oorschot [7] classify twenty-nine device-fingerprinting methods for augmenting web authentication and compare them by stability, spoofing difficulty, and distinguishability, and Andriamilanto *et al.* [8] show, on more than four million browsers, that browser fingerprints carry enough distinctiveness and stability to strengthen authentication. Freeman *et al.* [6] introduced the first public probabilistic model that scores a login as normal or suspicious from signals such as source IP, geolocation, browser configuration, and time of day. These works validate individual contextual signals but treat them largely in isolation and do not couple the resulting decision to a tamper-evident record.

Risk-based authentication. Wiefling *et al.* [4] conducted the first large-scale empirical study of RBA in the wild, and later work [5] showed that risk factors must be carefully tuned per service. Our proposal follows this line but replaces fixed heuristics—and the single-signal, statistically-scored models that preceded them [6]—with a multi-signal ML ensemble, and adds a non-repudiation layer.

Anomaly detection. Isolation Forest [14] is a reference for unsupervised outlier detection in high-dimensional spaces,

TABLE I
POSITIONING AGAINST REPRESENTATIVE RELATED WORK

Work	Signals	Adapt.	Audit	Data
TypeNet [11]	Keystroke	✓	—	Real
Touchalytics [13]	Touch	✓	—	Real
Freeman [6]	IP/geo/UA	Part.	—	Real
Alaca [7]	Device fp.	—	—	—
Andriam. [8]	Browser fp.	—	—	Real
RBA [4]	IP/context	Part.	—	Real
Iso. Forest [14]	Generic	Unsup.	—	—
BlockAudit [10]	Log events	—	✓	—
This work	14 ctx.+beh.	Ens.	✓	Synth.

Adapt.: adaptive/learned decision (Part.: partial; Ens.: ensemble). *Audit*: immutable audit coupled to the decision.

and supervised and deep models [16] complement it when labels are available; Killourhy and Maxion [15] established a methodological benchmark for keystroke anomaly detection. We combine both natures—unsupervised and supervised—in an ensemble.

Blockchain for auditing. Although blockchain-based log auditing has been investigated, most approaches treat event recording as *decoupled* from the decision mechanism. Systems such as BlockAudit [10] demonstrate scalable, tamper-evident audit logs on a blockchain, but the logged events originate in applications external to the recording layer, so the record attests *that* an event occurred without binding it to the reasoning that produced it. The distinguishing feature of this work is to integrate, in a single flow, the adaptive AI decision and its immutable record on a permissioned ledger, so that each stored entry carries the very features and score that justified the action. Table I summarizes representative work: no single line of research simultaneously provides (i) fusion of multiple contextual and behavioral signals, (ii) an adaptive ensemble decision, and (iii) audit immutability coupled to the decision.

III. PROPOSED ARCHITECTURE

The architecture articulates microservices, federated identity management, AI-based risk analysis, and immutable blockchain recording, following the principles of security by design and continuous verification of Zero Trust [3], [20]. Rather than a binary decision at login time, the model adopts a continuous, risk-oriented perspective. It is organized in five integrated layers (Fig. 1):

- 1) **Modular Application.** Independent services (Spring Boot) and an Angular front end exposing authenticated REST APIs.
- 2) **Identity and Access.** Federated authentication through Keycloak (OAuth2 / OpenID Connect), JWT issuance, and MFA support.
- 3) **Contextual Intelligence.** Extraction of the fourteen behavioral features and computation of the risk score by the ensemble.

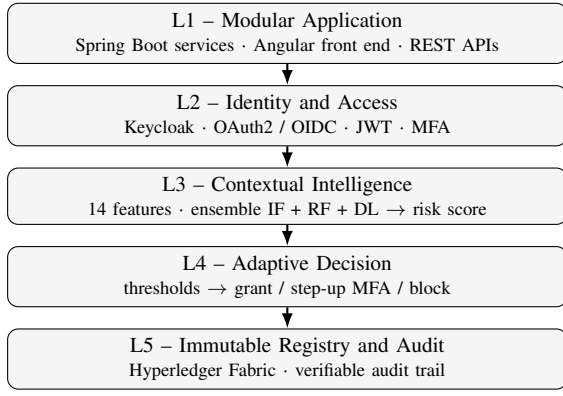


Fig. 1. Five-layer architecture of the proposed system.

- 4) **Adaptive Decision.** Application of the risk thresholds, resulting in *grant*, *step-up to MFA*, or *block*.
- 5) **Immutable Registry and Audit.** Persistence of each decision on a permissioned blockchain, ensuring traceability and non-repudiation.

The operational flow is a chained decision pipeline: a login attempt triggers extraction of the contextual features; the AI engine computes the risk score; the decision thresholds are applied; the outcome is recorded on the blockchain; and the system grants access, requires MFA, or blocks the attempt. The separation into layers allows each responsibility to evolve independently and specific security policies to be enforced at each level.

Threat model and design rationale. The system assumes an adversary who may already hold valid credentials—obtained through leakage, phishing, or reuse—and who can attempt logins from arbitrary networks and devices, but who cannot forge the consensus of the permissioned ledger nor observe the private state of the risk models. Under this model the layers address complementary concerns: L2 establishes *who* claims to authenticate, L3 estimates *how likely* that claim is legitimate given the context, L4 turns the estimate into a proportionate action, and L5 makes the action *accountable*. Decoupling estimation (L3) from enforcement (L4) is deliberate: risk thresholds can be re-tuned, and models retrained, without touching the identity provider, and the audit layer stays agnostic to the specific learners in use. Because each decision is written together with its justification—the hash of the input features and the score that produced it—an auditor can later replay any decision and verify that the recorded action matches the policy in force at that time, without having to trust the operator’s own logs. This tight coupling between *deciding* and *recording* is the property that distinguishes the proposal from pipelines that log events after, and separately from, the decision.

IV. AI DECISION ENGINE

Ensemble. The risk engine combines three complementary learners. *Isolation Forest* [14] acts as the first level of analysis, isolating outliers in the multidimensional feature space with

few random partitions; it contributes 40% of the final weight. *Random Forest* contributes 30%, capturing non-linear relations among features while remaining interpretable. A *multilayer neural network* [16] contributes the remaining 30%, with an input layer of 14 neurons, two hidden layers (64 and 32 units, ReLU activations, dropout regularization), and a sigmoid output. The final score is the weighted average

$$S = 0.4 S_{IF} + 0.3 S_{RF} + 0.3 S_{DL}, \quad (1)$$

where each component returns a value in $[0, 1]$ (0: normal, 1: highly anomalous). Combining unsupervised and supervised learners increases robustness both to novel (zero-day) and to known attacks.

Features. Each attempt is represented by fourteen features grouped into four families: *temporal* (access hour on a circular scale, weekday, time since last login, 24 h access frequency); *geographic/network* (geodesic distance from the usual location, network type, ASN, impossible-travel speed); *device/browser* (browser fingerprint, operating system, device type, automation/headless detection); and *session behavior* (keystroke pattern and recent failure history). These signals target credential stuffing, impossible travel, session hijacking, bots, and zero-day deviations.

Adaptive thresholds. The score is mapped to three risk bands, empirically calibrated to balance security and usability: low risk ($S < 0.3$) grants access without additional steps; moderate risk ($0.3 \leq S < 0.7$) triggers MFA; and high risk ($S \geq 0.7$) blocks the attempt, records the event on the blockchain, and notifies the security administrator. In production, the engine is exposed as an independent REST microservice consumed by the back end, keeping the three models in memory for parallel inference. A simplified heuristic fallback preserves availability if the AI service is unreachable. Models are periodically retrained to track behavioral change (concept drift) [17], and every trained model is versioned to allow rollback.

V. BLOCKCHAIN-BASED AUDITING

Every decision is recorded immutably on a permissioned ledger based on Hyperledger Fabric [18], whose identity control and structured governance suit corporate and government environments [19]. For each event the ledger stores the hash of the input features, the risk score, the decision (grant, MFA, or block), and the transaction timestamp, chained to the previous record so as to form a chronological chain of evidence.

The choice of a blockchain over simpler mechanisms is justified rather than assumed. Signed logs and append-only databases already provide integrity and non-repudiation through cryptographic signatures, and are sufficient when a single trusted authority operates the registry. The essential difference of a permissioned blockchain is *not* the signature itself, but not depending on trust in a single custodian: the record is replicated and validated by consensus among multiple participants, becoming resistant to tampering even by privileged administrators of the system and enabling independent verification by external auditors. This is decisive here, because

access decisions are made autonomously by AI models and must be auditable by third parties (e.g., LGPD [21] compliance checks), a scenario in which the operator alone should not be able to alter the decision trail. We acknowledge the cost—higher operational complexity and lower throughput than an append-only store—which is warranted precisely in multi-party, regulated settings, and delimits the scope in which the blockchain is actually necessary.

VI. EVALUATION METHODOLOGY

Dataset and labeling. In the absence of public labeled datasets for contextual authentication, a synthetic dataset of about 50,000 login records was built,¹ reproducing the distribution observed in production (about 95% legitimate and 5% suspicious accesses). Each record contains the fourteen features of Section IV. Labels were derived from heuristic rules encoding known fraud patterns (impossible travel, unknown device, bursts of failed logins) and reviewed by security specialists. The generative process draws each legitimate record from distributions calibrated to typical corporate usage: the access hour follows a Gaussian centred on mid-day, the inter-login interval an exponential law, the 24-hour frequency a Poisson count, and the geodesic distance a half-normal concentrated near the habitual location; the browser-fingerprint and keystroke-similarity scores are high-mean Gaussians, while the device, operating-system, and network-type indicators follow the categorical frequencies observed in production. Suspicious records are obtained by perturbing two to five features of an otherwise legitimate profile into anomalous ranges — impossible-travel speeds, unknown devices, automation flags, and bursts of recent failures— so that anomalies remain multivariate and partially overlap the legitimate class instead of being trivially separable. A calibrated amount of Gaussian noise is then added to every feature to reproduce the class overlap expected in the wild, and about 5% of the legitimate records receive a mild single-feature anomaly, which yields the realistic false positives reflected in the reported FPR. Because the data are synthetic, no personally identifiable information is collected at this stage; the move to real data will follow the minimization and anonymization principles required by the LGPD [21].

Protocol. The dataset was split in a stratified way into 70% training and 30% testing. Stratified 5-fold cross-validation was used on the training partition for hyperparameter selection, keeping the test partition exclusively for the final evaluation to avoid data leakage. Performance was measured by accuracy, precision, recall, F1-score, false-positive rate (FPR), ROC-AUC, and decision latency. Random seeds were fixed, models and hyperparameters were versioned, and the whole environment was containerized (Docker/Kubernetes) to allow deterministic reproduction. To support independent verification, the source code, the trained-model artifacts, and the scripts

that generate the dataset and reproduce every experiment are openly available.²

Reproducibility. A single entry-point script regenerates the entire pipeline deterministically from the fixed seed: the labeled dataset (50,000 rows with the fourteen features and the class label), the three trained learners together with the feature scaler (serialized with `joblib`), the ensemble configuration (component weights, the Isolation-Forest score-normalization bounds, and the decision threshold selected by the best training-set F1), and a machine-readable metrics file holding every figure reported in Section VII. Re-executing the script on a clean environment reproduces the reported metrics up to the negligible variation introduced by the stochastic optimizer of the neural network, so that a reviewer can audit the whole pipeline end to end—from raw feature generation to the final decision threshold—without access to any private or personally identifiable data. This packaging is what turns the “about 50,000 records” description into an artifact that a third party can regenerate, inspect, and extend.

Baselines. Table II contrasts the proposal with representative baselines from the literature and practice. The baseline figures are taken from the cited literature and were not re-executed here; the comparison is therefore qualitative regarding the capabilities of each approach. The three comparison axes are not arbitrary; they follow from the requirements that the literature places on modern authentication. *Context-awareness* (Ctx.) reflects the NIST and Zero-Trust prescription that verification be conditioned on the signals of each attempt rather than on the credential alone [1], [3], and is the property that empirical studies of risk-based authentication find decisive in the wild [4], [6]. *Adaptivity* (Adapt.) captures whether the response is proportional to the estimated risk—the core principle of risk-based authentication [1], [4], [5]—rather than a fixed, one-size-fits-all rule. *Auditability* (Audit.) addresses the accountability of automated decisions over personal data, a requirement made explicit by data-protection regulation such as the LGPD [21] and, more broadly, by the need for independent verification when access is delegated to autonomous models. Together these axes span the two properties a contextual-authentication system must jointly satisfy—deciding correctly and being verifiable—while the *main limitation* column indicates where each baseline falls short on at least one of them.

VII. RESULTS AND DISCUSSION

On the held-out test set (30% of the data, unseen during training) the ensemble reached an accuracy of 98.1%, precision of 83.0%, and recall of 77.6% for suspicious accesses, with an F1-score of 80.2% and a ROC-AUC of 0.99, indicating strong ranking capability between legitimate and suspicious accesses. The false-positive rate remained low at 0.8%, which is relevant to usability. Stratified 5-fold cross-validation confirmed the stability of the results (accuracy $98.2\% \pm 0.1\%$; F1 $80.5\% \pm 0.8\%$). Table III summarizes these figures.

¹Dataset (labeled CSV, trained models, and metrics): <https://github.com/jgbcientista/gestao-autonomo/tree/main/docs/artefatos-modelos>

²<https://github.com/jgbcientista/gestao-autonomo>

TABLE II
COMPARISON WITH REFERENCE APPROACHES

Approach	Ctx.	Adapt.	Audit.	Main limitation
Password static 2FA	+	—	—	Ignores context; phishing/SIM swap
IP-based RBA [4]	Part.	Heur.	—	Single feature; manual tuning
Statistical scoring [6]	✓	Stat.	—	Few signals; no decision audit
Keystroke only [15]	✓	—	—	Offline, single factor (EER $\approx 9.6\%$)
Proposed	✓	Ens.	✓	Adaptive, explainable, auditable

Ctx.: uses login context; Adapt.: adaptive decision (Heur.: heuristic; Stat.: statistical; Ens.: ensemble); Audit.: immutable audit coupled to the decision.

TABLE III
ENSEMBLE PERFORMANCE ON THE HELD-OUT TEST SET

Metric	Value	Metric	Value
Accuracy	98.1%	F1-score	80.2%
Precision	83.0%	ROC-AUC	0.99
Recall	77.6%	False-pos. rate	0.8%
5-fold accuracy	98.2% $\pm$ 0.1%		
5-fold F1-score	80.5% $\pm$ 0.8%		
Ensemble latency	≈ 33 ms/request		

Regarding the authentication chain, the integration between Keycloak and the ML service was validated successfully. Ensemble inference was measured at about 33 ms per request on the reference machine (median of single-request calls, dominated by the Random Forest), well below the 200 ms budget set for the AI component; the complete contextual authentication flow—including identity validation and blockchain recording—stayed in the order of a few hundred milliseconds. The blockchain pipeline was consistent under the immutability tests, with successful cryptographic validation of every recorded transaction, and ELK Stack and Grafana provided real-time visualization of the authentication events.

Discussion. The partial results are promising and suggest the technical feasibility of the approach; however, because they stem from synthetic data and a preliminary stage, they do not yet allow the central hypothesis to be confirmed conclusively. A rigorous comparison against the baselines and validation with real data remain necessary before any quantitative claim of superiority. The F1-score of 80.2%, although adequate for the current stage, indicates room to improve recall once real data replace the synthetic set. Even so, in a controlled environment the adaptive, context-driven approach detected suspicious access patterns—such as the simulated credential-stuffing and session-hijacking scenarios—that escape systems based solely on passwords and static 2FA, while the blockchain layer made retroactive tampering of the records infeasible without

detection.

Error analysis and trade-offs. The decision threshold is not fixed a priori but selected as the value that maximizes the F1-score on the training partition, which makes the operating point explicit and tunable per deployment: lowering it raises recall at the cost of more step-up challenges, whereas raising it favours a frictionless experience at the risk of missing borderline attacks. At the chosen point the confusion matrix on the test set is dominated by true negatives, and the residual errors concentrate on the boundary between mildly anomalous legitimate accesses and low-intensity suspicious ones—exactly the region the noise model was designed to make ambiguous. This explains both the sub-90% F1 and the low false-positive rate: the ensemble is conservative, preferring an occasional missed detection over penalizing legitimate users, a bias that is appropriate for an authentication front-end where usability friction carries a direct business cost. The three learners contribute complementary views—the Isolation Forest flags globally rare feature combinations without labels, while the Random Forest and the neural network exploit the labeled decision boundary—so that the weighted average degrades gracefully when any single learner is uncertain. We deliberately refrain from reading these figures as evidence of superiority over the baselines of Table II: the baselines were not re-executed on the same data, and the synthetic generator, however carefully calibrated, cannot capture the full adversarial diversity of production traffic. These observations frame the experiment as a feasibility study whose next step is validation on real, non-repudiable logs collected under the very architecture proposed here.

Limitations and threats to validity. Three limitations bound the reach of our conclusions. First, external validity is constrained by the use of synthetic data: although the generative process reproduces plausible corporate-usage distributions, it captures neither the long tail of atypical yet legitimate behavior nor the adversarial creativity observed in real traffic, so the reported metrics should be read as an optimistic upper bound rather than as expected production performance. Second, construct validity depends on heuristic labeling: because the labels derive from the same rules that describe the fraud patterns, they may introduce a circularity in which the ensemble re-learns the boundaries encoded in the generator itself—a risk that only labeling over real, analyst-reviewed incidents fully removes. Third, the comparison against the baselines is deliberately qualitative, as they were not re-executed on the same dataset; quantitative claims of superiority therefore remain outside the scope of this preliminary phase. We partially mitigate these threats by fixing random seeds, versioning models and hyperparameters, and openly publishing the artifacts for independent inspection; full mitigation, however, requires collecting real, non-repudiable logs under the proposed architecture—precisely the immutable record the blockchain audit layer provides—closing the loop between methodological evaluation and actual system operation.

VIII. CONCLUSION AND FUTURE WORK

This paper presented an autonomous contextual authentication system that unifies, in a single flow, an adaptive AI decision and its immutable audit on a permissioned blockchain. The main contributions are the end-to-end architecture fusing fourteen contextual and behavioral signals through an ensemble, the decision-coupled audit layer with an explicit justification for the use of blockchain, and a reproducible evaluation methodology. Preliminary experiments on synthetic data indicated technical feasibility (98.1% accuracy, 80.2% F1-score, 0.99 AUC, ≈ 33 ms inference latency), constituting a methodological validation rather than a substitute for evaluation on real data. Future work includes replacing the synthetic dataset with real or semi-real data collected in a controlled environment, refining model and threshold calibration, assessing performance and scalability under realistic load, and deepening the security analysis against the threat model. We further plan to investigate the explainability of the ensemble decisions—exposing, for each block or MFA challenge, the criteria that contribute most to the risk score—to evaluate system behavior under prolonged concept drift and adversarial evasion attacks, and to conduct a usability study that quantifies the friction perceived by legitimate users when facing adaptive challenges. On the audit side, we intend to measure the consensus cost of the permissioned network under realistic login rates and to evaluate periodic anchoring strategies that preserve independent verifiability without compromising throughput. We also plan to extend the ensemble with online learning so that model updates track behavioral drift continuously rather than only at periodic retraining boundaries, and to study calibration techniques that keep the risk score aligned with the empirical probability of compromise as the operating environment evolves. Finally, integrating the proposed decision-and-audit pipeline with a federated identity deployment at institutional scale would let us assess the end-to-end latency budget, the operational overhead of key and certificate management, and the governance model required for multiple organizations to jointly operate and audit the permissioned ledger. Taken together, these directions aim to progressively close the gap between the controlled feasibility study reported here and a production-grade deployment whose adaptive decisions remain, at every step, both effective and independently verifiable.

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